Lecture 15

Hydrogen: the spectrum

Objectives. *Lecture 14 proved that the Hamiltonian H of hydrogen is self-adjoint, that its spectrum lies in $[-\mathrm{Ry}, \infty)$ and contains $[0, \infty)$, and that every negative spectral point is an eigenvalue. We find these eigenvalues, prove their multiplicities and compare their transitions with the Balmer lines.*

- Construct the bound eigenvectors of energy $-\mathrm{Ry}/n^2$.
- Prove that their eigenspaces have dimension exactly n^2.
- Compute Balmer wavelengths and distinguish rotational from additional degeneracy.

15.1 The bound states

Lecture 14 proved that the Hamiltonian of hydrogen is self-adjoint (Theorem 14.2),

$$H = H_0 - \frac{C}{|x|} \ \text{ on } D(H) = D(H_0), \qquad a_0 = \frac{\hbar^2}{mC}, \qquad \mathrm{Ry} = \frac{\hbar^2}{2ma_0^2} \approx 13.61\,\mathrm{eV},$$

and proved what else this lecture uses. Every vector of $D(H_0)$ is a bounded continuous function (Lemma 14.1). A bounded continuous function ψ that is square integrable, twice continuously differentiable on $x \neq 0$, with bounded first partial derivatives there and a square-integrable $\Delta\psi$, lies in $D(H_0)$, and $H_0\psi = -(\hbar^2/2m)\Delta\psi$ (Lemma 14.3). Every $\psi \in D(H)$ has $\langle \psi, H\psi\rangle \geq -\mathrm{Ry}\,\|\psi\|^2$, and $\psi_1(x) = e^{-|x|/a_0}$ is an eigenvector for $-\mathrm{Ry}$ (Theorem 14.4). The essential spectrum of H is $[0, \infty)$, and every point of $\sigma(H)$ below 0 is an eigenvalue in $\sigma_{\mathrm{d}}(H)$ (Proposition 14.5 and Theorem 14.7). Every eigenvalue is negative (Proposition 14.8). What remains is to find the eigenvalues and their eigenvectors.

Constructing the bound states

Below 0 we look for eigenvectors of the form of Proposition 13.11. For $p \in \mathcal{V}_l$, the function $\psi = u(|x|)p(x)/|x|^{l+1}$ satisfies $-(\hbar^2/2m)\Delta\psi + V\psi = E\psi$ on $x \neq 0$ when $H_l u = Eu$, with $V(r) = -C/r$ in the radial rule H_l. Write $E = -\hbar^2\eta^2/2m$ with $\eta > 0$. Far out the equation is close to $u'' = \eta^2 u$, whose decaying solution is $e^{-\eta r}$, and near 0 the centrifugal term makes $u \approx r^{l+1}$. So put

$$u(r) = r^{l+1}e^{-\eta r}v(r).$$

By the product rule,

$$\frac{u''}{r^{l+1}e^{-\eta r}} = v'' + 2\Big(\frac{l+1}{r} - \eta\Big)v' + \Big(\frac{l(l+1)}{r^2} - \frac{2\eta(l+1)}{r} + \eta^2\Big)v.$$

Dividing $H_l u = Eu$ by $-\hbar^2/2m$, with $2mC/\hbar^2 = 2/a_0$, gives $u'' = (l(l+1)/r^2 - 2/a_0 r + \eta^2)u$. So $H_l u = Eu$ exactly when

$$rv'' + 2(l+1-\eta r)v' + 2\Big(\frac{1}{a_0} - \eta(l+1)\Big)v = 0. \tag{15.1}$$

For a series $v = \sum_k c_k r^k$, the coefficient of r^k gives

$$(k+1)(k+2l+2)\,c_{k+1} = 2\Big(\eta(k+l+1) - \frac{1}{a_0}\Big)c_k. \tag{15.2}$$

The series stops after the degree n_r exactly when $\eta(n_r + l + 1) = 1/a_0$. Then $\eta = 1/na_0$ with the *principal quantum number* $n = n_r + l + 1$, and $E = -\hbar^2/2mn^2a_0^2 = -\mathrm{Ry}/n^2$. The series must stop: the fourth step of the proof of Theorem 15.4 shows that a series that does not stop grows so fast that u is not square integrable.

Theorem 15.1 (The levels of hydrogen). *Let $n \geq 1$ and $0 \leq l \leq n-1$ be integers, and let $v(r) = \sum_{k=0}^{n-l-1} c_k r^k$ with*

$$c_0 = 1, \qquad c_{k+1} = \frac{2(k+l+1-n)}{(k+1)(k+2l+2)\,na_0}\,c_k.$$

For every $p \in \mathcal{V}_l$, the function $\psi(x) = p(x)\,v(|x|)\,e^{-|x|/na_0}$ lies in $D(H)$, and

$$H\psi = -\frac{\mathrm{Ry}}{n^2}\,\psi.$$

The eigenspace of H for $-\mathrm{Ry}/n^2$ has dimension at least n^2.

Proof.

1. *The radial equation.* Let $\eta = 1/na_0$, so that $\eta(k+l+1) - 1/a_0 = (k+l+1-n)/na_0$, and put $c_k = 0$ for $k \geq n-l$. Then the c_k satisfy (15.2) for every $k \geq 0$, since for $k = n-l-1$ its right side vanishes. So the polynomial v satisfies (15.1) in every power of r, and $u = r^{l+1}e^{-\eta r}v$ has $H_l u = -(\mathrm{Ry}/n^2)u$.
2. *An eigenvector.* Since $\psi = up/r^{l+1}$, Proposition 13.11 gives $-(\hbar^2/2m)\Delta\psi = -(V + \mathrm{Ry}/n^2)\psi$ on $x \neq 0$. The function ψ, a polynomial in x and $|x|$ times $e^{-|x|/na_0}$, is bounded, continuous and square integrable. On $x \neq 0$ it is smooth, and its first partial derivatives are bounded, since $\partial_j |x| = x_j/|x|$ is bounded. Finally $V\psi$ is square integrable, since ψ is bounded and square integrable and $|x|^{-2}$ is integrable near 0, and so is $\Delta\psi$. By Lemma 14.3, $\psi \in D(H_0) = D(H)$ and $H_0\psi = -(\hbar^2/2m)\Delta\psi$. So $H\psi = -(\mathrm{Ry}/n^2)\psi$.
3. *The dimension.* Each $\mathcal{V}_l$ has dimension $2l+1$ (§ 13.4). Let $\varphi_l = p_l v_l e^{-|x|/na_0}$, with $p_l \in \mathcal{V}_l$ and v_l the polynomial for l, and suppose that $\sum_{l=0}^{n-1} \varphi_l = 0$. On $x \neq 0$ the L_j commute with functions of $|x|$ (Proposition 13.4), and $L^2 p_l = \hbar^2 l(l+1)p_l$, so $L^2\varphi_l = \hbar^2 l(l+1)\varphi_l$. Applying the product of the $L^2 - \hbar^2 l'(l'+1)$ over $l' \neq l$ to the sum leaves φ_l times a number that is not 0, since the $l(l+1)$ differ. So $\varphi_l = 0$. Since $v_l(0) = 1$, $v_l \neq 0$ near 0, so p_l vanishes near 0, and the polynomial p_l is 0. Hence the dimension is at least $\sum_{l=0}^{n-1}(2l+1) = n^2$. ∎

Example 15.2 (The level $n = 2$). For $n = 2$ and $l = 0$, the recursion of Theorem 15.1 gives $c_1 = -c_0/2a_0$, so $v = 1 - r/2a_0$. For $n = 2$ and $l = 1$, the series stops at once, so $v = 1$. So the level $-\mathrm{Ry}/4$ holds

$$\Big(1 - \frac{|x|}{2a_0}\Big)e^{-|x|/2a_0} \quad \text{and} \quad x_j e^{-|x|/2a_0}, \quad j = 1, 2, 3.$$

The first function vanishes on the sphere $|x| = 2a_0$. These four functions give the bound $n^2 = 4$ of Theorem 15.1.

For $n = 1$, $v = 1$ and $\psi = \psi_1$ of Theorem 14.4. The polynomials v are Laguerre polynomials (Teschl, (10.65)).

Normalise $u = r^{l+1}e^{-r/na_0}v$ so that $\int_0^\infty |u|^2 \, dr = 1$, and normalise $Y = p/r^l$ on the sphere. Then the state $\psi = (u/r)Y$ is a unit vector, and $\int_{a_1}^{b_1} |u|^2 \, dr$ is the probability that a reading of the radius $|x|$ lies in (a_1, b_1), as in § 13.4. For $n = 1$ this is Example 13.9 with $a = a_0$, so the probabilities 0.323 and 0.439 found there are those of the state of ψ_1. Figure 15.1 draws $|u|^2$ for $n \le 3$, with the polynomials v of the recursion. A zero of v at $r > 0$ is a *node sphere*: the eigenvector vanishes on a whole sphere. The state with $n = 2$ and $l = 0$ has the node sphere $|x| = 2a_0$ of Example 15.2.

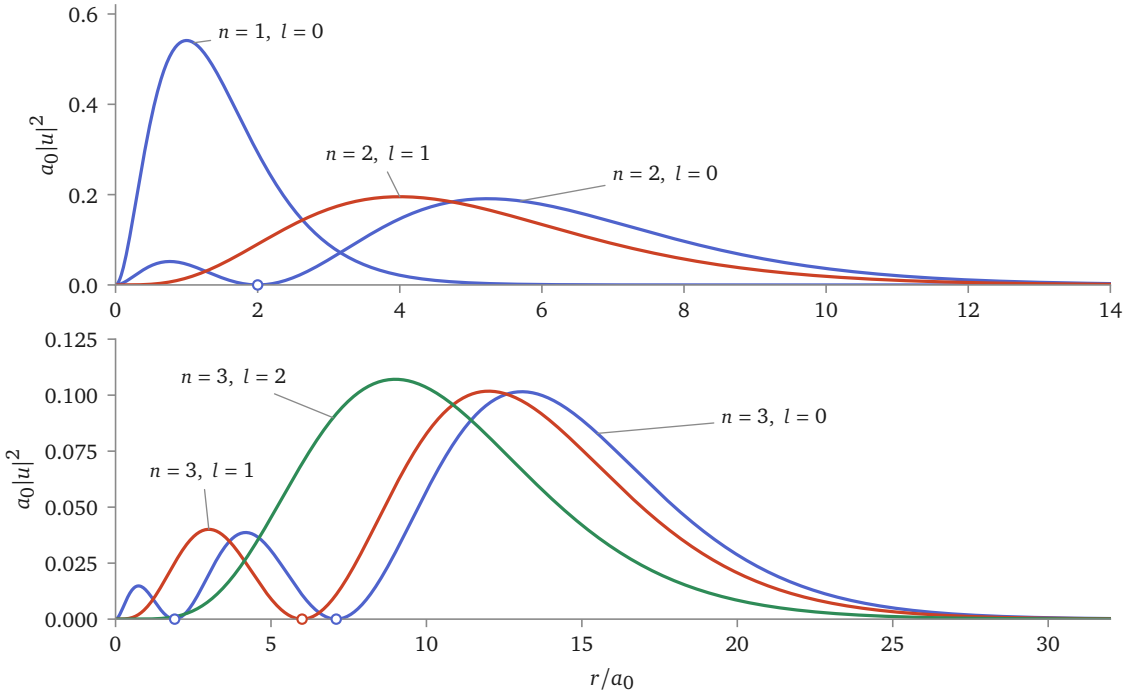


Figure 15.1: The density $a_0\,|u|^2$ of the scaled radius r/a_0, for the levels $n = 1, 2$ (above) and $n = 3$ (below), with one colour for each l. The area under a curve between two radii is the probability of that shell. A hollow mark is a node sphere. A curve shows this density of the radius, not the spatial density $|\psi|^2$: for $l = 0$ the curve vanishes at $r = 0$, and the spatial density does not.

The radial rule H_l of Proposition 13.11 is that of a particle on the half-line in the potential $\hbar^2 l(l+1)/2mr^2 - C/r$, which equals $\mathrm{Ry}\,(l(l+1)a_0^2/r^2 - 2a_0/r)$. Figure 15.2 draws it for $l = 0, 1, 2$, with the levels of Theorem 15.1. For $l = 0$ the potential tends to $-\infty$ at the centre. For $l \ge 1$ the centrifugal term makes it tend to $+\infty$ there. The panel of l shows the levels with $n \ge l + 1$, which hold states of this l by Theorem 15.1. That the other levels hold none follows from the termination of the series, and Theorem 15.4 below proves it.

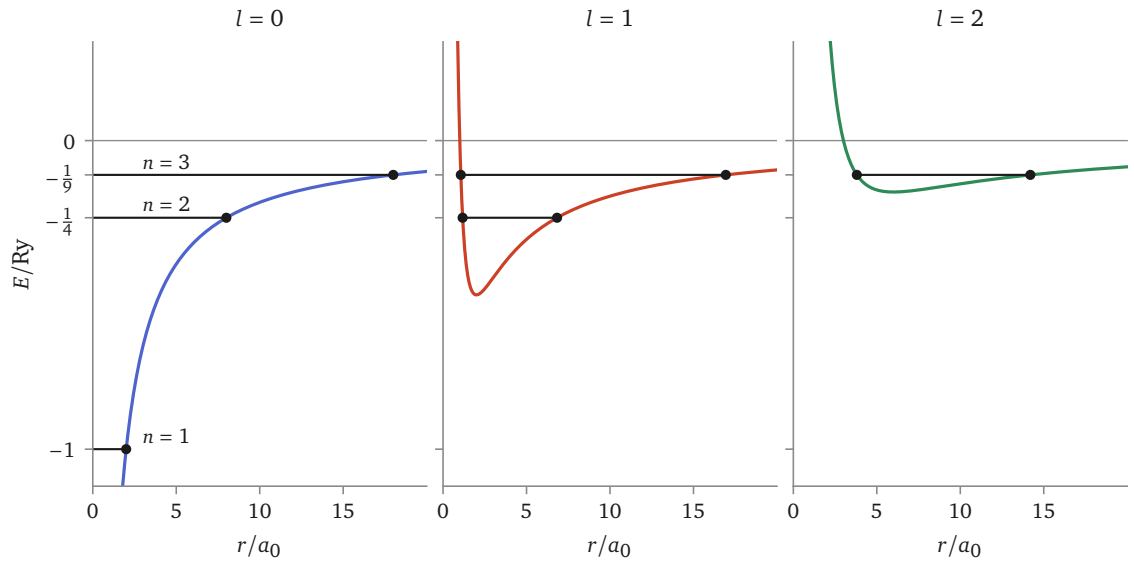


Figure 15.2: The potential $l(l+1)(a_0/r)^2 - 2a_0/r$ of the radial rule, in units of Ry, against the scaled radius r/a_0, for $l = 0, 1, 2$, one panel each, in the colours of Figure 15.1. A horizontal segment is a level $-1/n^2$ with $n \le 3$ and $n \ge l+1$, drawn where it lies above the curve, and a dot is a turning point. For $l = 0$ the segment reaches the centre. The labels in the panel of $l = 0$, where every level is drawn, name the levels.

A classical particle with the energy $-\mathrm{Ry}/n^2$ and an angular momentum whose square is $\hbar^2 l(l+1)$ moves only where the level lies above the curve. For $l \ge 1$ it turns back at the radii $a_0\bigl(n^2 \pm n\sqrt{n^2 - l(l+1)}\bigr)$, and for $l = 0$ it stays in $0 < r \le 2n^2 a_0$. Each density of Figure 15.1 is largest inside this range for its n and l, and it does not vanish outside it, as for the oscillator in Figure 8.1. For $n = 1$, the primitive of Example 13.9 gives $\mathrm{Prob}(|x| > 2a_0) = 13e^{-4} \approx 0.238$ in the state of ψ_1.

Completeness and multiplicity

Theorem 15.1 constructs, at each level $-\mathrm{Ry}/n^2$, eigenvectors of one form, $p(x)v(|x|)e^{-|x|/na_0}$ with $p \in \mathcal{V}_l$, and so at least n^2 independent ones. It does not show that every eigenvector is a combination of these, nor that no other negative number is an eigenvalue. Proposition 14.8 already excludes the numbers at or above 0. Theorem 15.4 below proves both remaining claims. Let ψ be an eigenvector of H. By Lemma 14.1, ψ is continuous. For $p \in \mathcal{V}_l$, its *radial coefficient* along p is

$$u_p(r) = r \int_{S^2} \overline{p(\hat{x})}\, \psi(r\hat{x})\, \mathrm{d}\Omega(\hat{x}), \qquad r > 0,$$

where $\hat{x}$ runs over the unit sphere S^2, with $\mathrm{d}\Omega = \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\varphi$ (§ 13.4). The radial coefficient u_p depends antilinearly on p. A lemma from calculus will turn the eigenvalue equation into the radial equation for u_p, and a fact about polynomials will show that ψ is determined by its radial coefficients.

Lemma 15.3 (Weak solutions). *Let u and f be continuous on an open interval J, and suppose that $\int_J uq''\,\mathrm{d}r = \int_J fq\,\mathrm{d}r$ for every smooth q that vanishes outside a closed bounded interval inside J. Then u is twice continuously differentiable, and $u'' = f$.*

Proof. Fix $r_0 \in J$ and let $\Phi(r) = \int_{r_0}^{r} \int_{r_0}^{t} f(s)\,\mathrm{d}s\,\mathrm{d}t$, so that $\Phi'' = f$ and, integrating by parts twice, $\int \Phi q'' = \int f q$. So $w = u - \Phi$ has $\int w q'' = 0$ for every such q. Choose two such functions q_0 and q_1 with $\int q_0 = 1$, $\int r q_0 = 0$, $\int q_1 = 0$ and $\int r q_1 = 1$ (combinations of a bump and its translate). For every such q, the function $\tilde{q} = q - (\int q) q_0 - (\int r q) q_1$ has $\int \tilde{q} = \int r\tilde{q} = 0$. So $y(r) = \int_a^r \int_a^t \tilde{q}(s)\,\mathrm{d}s\,\mathrm{d}t$, with a below the supports, vanishes outside a closed bounded interval and has $y'' = \tilde{q}$, and $\int w\tilde{q} = \int w y'' = 0$. Hence $\int wq = A\int q + B\int rq$, that is, $\int (w - A - Br) q = 0$ for every q, with $A = \int w q_0$ and $B = \int w q_1$. A continuous function with this property vanishes, since otherwise its real or imaginary part has one sign on a small interval, where a bump q gives a nonzero integral. So $u = \Phi + A + Br$, which is twice continuously differentiable, with $u'' = \Phi'' = f$. ∎

Every polynomial is a sum of terms $|x|^{2j}\,p(x)$ with $p \in \mathcal{V}_l$. Indeed, let $N \geq 2$, and recall the Gaussian $g(x) = e^{-|x|^2/2s^2}$ of Lecture 13, for a fixed length $s > 0$. Multiplication by $|x|^2$ commutes with the L_j (Proposition 13.4) and maps $\mathcal{P}_{N-2}$ one-to-one into $\mathcal{P}_N$, so L^2 has on $|x|^2\,\mathcal{P}_{N-2}g$ the eigenvalues $\hbar^2 l(l+1)$, $l = N-2, N-4, \ldots$ (§ 13.4), and on $\mathcal{V}_N g$ the eigenvalue $\hbar^2 N(N+1)$. The two spaces meet only in 0, and their dimensions add up to that of $\mathcal{P}_N$, so $\mathcal{P}_N = \mathcal{V}_N \oplus |x|^2\,\mathcal{P}_{N-2}$. Since $\mathcal{P}_0 = \mathcal{V}_0$ and $\mathcal{P}_1 = \mathcal{V}_1$, induction on N gives the claim.

Theorem 15.4 (The spectrum of hydrogen)**.** *The spectrum of H is*

$$\sigma(H) = \left\{ -\frac{\mathrm{Ry}}{n^2} : n = 1, 2, \ldots \right\} \cup [0, \infty), \qquad \sigma_{\mathrm{ess}}(H) = [0, \infty).$$

Every eigenvalue of H is one of the $-\mathrm{Ry}/n^2$, and its eigenspace has dimension n^2. It is spanned by the functions of Theorem 15.1.

Proof.

1. *The radial coefficients.* Let $H\psi = E\psi$ with $\psi \neq 0$. By Proposition 14.8, $E = -\hbar^2\eta^2/2m$ with $\eta > 0$. Let $p \in \mathcal{V}_l$. Since ψ is continuous and bounded, u_p is continuous, and $|u_p(r)| \leq r \sup|\psi| \int_{S^2} |p|\,\mathrm{d}\Omega$. By the Cauchy–Schwarz inequality on the sphere, $\int_0^\infty |u_p|^2\,\mathrm{d}r \leq \int_{S^2} |p|^2\,\mathrm{d}\Omega\,\|\psi\|^2$.
2. *The radial equation.* Let q be smooth and vanish outside a closed bounded interval inside $(0,\infty)$, and put $\zeta(x) = q(|x|)p(x)/|x|^{l+1}$. By Proposition 13.11, $\zeta \in \mathcal{S}(\mathbb{R}^3)$ and $H\zeta = (H_l q)(|x|)p(x)/|x|^{l+1}$. Integrating in spherical coordinates, $\langle \zeta, \psi\rangle = \int_0^\infty \bar{q} u_p\,\mathrm{d}r$, and in the same way $\langle H\zeta, \psi\rangle = \int_0^\infty \overline{H_l q}\,u_p\,\mathrm{d}r$. Since $\langle H\zeta, \psi\rangle = \langle \zeta, H\psi\rangle = E\langle \zeta, \psi\rangle$, dividing by $-\hbar^2/2m$ gives, for real q, and so by linearity for every q,
$$\int_0^\infty q'' u_p\,\mathrm{d}r = \int_0^\infty q\left(\frac{l(l+1)}{r^2} - \frac{2}{a_0 r} + \eta^2\right) u_p\,\mathrm{d}r.$$
By Lemma 15.3, u_p is twice continuously differentiable and solves the radial equation $H_l u_p = E u_p$.
3. *The regular solution.* Let $c_0 = 1$ and define the c_k by (15.2). Either the series stops, or every c_k is nonzero and $c_{k+1}/c_k \to 0$. In both cases $v = \sum_k c_k r^k$ converges for every r, can be differentiated term by term, and $u_1 = r^{l+1}e^{-\eta r}v$ solves the radial equation. The Wronskian $W = u_1 u_p' - u_1' u_p$ is constant, since $W' = u_1 u_p'' - u_1'' u_p = 0$. Since $u_1(r)/r^{l+1} \to v(0) = 1$ as $r \to 0$, we can choose $\epsilon \in (0,1)$ with $r^{l+1}/2 \leq u_1(r) \leq 2r^{l+1}$ on $(0,\epsilon]$. Suppose $W \neq 0$. On $(0,\epsilon]$, $(u_p/u_1)' = W/u_1^2$, so
$$u_p(r) = u_1(r)\left(\frac{u_p(\epsilon)}{u_1(\epsilon)} - W\int_r^\epsilon \frac{\mathrm{d}s}{u_1(s)^2}\right), \qquad \int_r^\epsilon \frac{\mathrm{d}s}{u_1(s)^2} \geq \frac{r^{-2l-1} - \epsilon^{-2l-1}}{4(2l+1)},$$

and $|u_p(r)| \geq b\, r^{-l}$ for small r, with some $b > 0$. This contradicts the bound $|u_p(r)| \leq \text{const} \cdot r$ of the first step. So $W = 0$. Let $\beta = u_p(\epsilon)/u_1(\epsilon)$. The solution $w = u_p - \beta u_1$ has $w(\epsilon) = 0$ by the choice of β, and its Wronskian with u_1 is $u_1 w' - u_1' w = W = 0$. At ϵ this gives $u_1(\epsilon) w'(\epsilon) = 0$, and $u_1(\epsilon) > 0$, so $w'(\epsilon) = 0$. Hence $w = 0$ by the proof of Lemma 10.5 on each closed bounded interval inside $(0, \infty)$. So $u_p = \beta u_1$.

4. *Square integrability.* If the series does not stop, every c_k is nonzero, and $(k+1)c_{k+1}/c_k \to 2\eta$. So for $k \geq K$ the c_k have one sign and $|c_{k+1}| \geq (3\eta/2)\,|c_k|\,/(k+1)$. Then

$$|v(r)| \geq |c_K|\; K! \left(\frac{3\eta}{2}\right)^{-K} \left(e^{3\eta r/2} - \sum_{k<K} \frac{(3\eta r/2)^k}{k!}\right) - \left|\sum_{k<K} c_k r^k\right|,$$

and u_1 grows at least like $e^{\eta r/2}$, so it is not square integrable. Since u_p is, $u_p = 0$ unless the series stops, that is, unless $\eta = 1/na_0$ with an integer $n \geq l+1$.

5. *All coefficients zero.* Suppose $u_p = 0$ for every l and every $p \in \mathcal{V}_l$. For $p \in \mathcal{V}_l$ and $j \geq 0$, with the Gaussian g of Lecture 13, integrating in spherical coordinates, $\left\langle |x|^{2j}\, pg, \psi \right\rangle = \int_0^\infty r^{2j+l+1} e^{-r^2/2s^2} u_p(r)\, \mathrm{d}r = 0$. By the decomposition of polynomials above, ψ is orthogonal to every $Q_N g$, so to every $h_{n_1 n_2 n_3}$, and $\psi = 0$ by Theorem 13.6.
6. *The eigenvalues.* If E is not one of the $-\mathrm{Ry}/n^2$, then by the fourth step every $u_p = 0$, and $\psi = 0$ by the fifth. So every eigenvalue is one of the $-\mathrm{Ry}/n^2$. For $E = -\mathrm{Ry}/n^2$, the fourth step gives $u_p = 0$ for $l \geq n$, and for $l \leq n-1$ the third gives $u_p = \beta(p) u_{nl}$ with $u_{nl} = r^{l+1} e^{-r/na_0} v$ of Theorem 15.1. Choose a basis $(p_{l,i})$ of each $\mathcal{V}_l$, $l \leq n-1$. By the fifth step, the linear map $\psi \mapsto (\beta(p_{l,i}))_{l,i}$ from the eigenspace to $\mathbb{C}^{n^2}$ is one-to-one. So the eigenspace has dimension at most n^2, and by Theorem 15.1 exactly n^2, spanned by the functions found there.
7. *The spectrum.* By Theorem 14.7, every point of $\sigma(H)$ below 0 is an eigenvalue, and every $-\mathrm{Ry}/n^2$ is one. With Proposition 14.5 and Theorem 14.7, this is the claim. ∎

Lecture 13 asked which self-adjoint operator on $L^2(0,\infty)$ the Hamiltonian becomes on the functions $(u(r)/r)Y_{l\mu}$, and on which domain. The proof above does not need the answer, which we state.

Theorem 15.5 (The sectors · *cited*). *Let H be H_0, or $H_0 - C/|x|$ with $C > 0$, on $D(H_0)$. For each l and μ, the projection onto the vectors $(u(r)/r)Y_{l\mu}$ maps $D(H)$ into itself and commutes with H, and in terms of u, H acts on these vectors as a self-adjoint operator A_l on $L^2(0,\infty)$: $A_l u = H_l u$ on the functions $u \in L^2(0,\infty)$ with u and u' absolutely continuous on every closed bounded interval inside $(0,\infty)$ and $H_l u \in L^2(0,\infty)$, and, for $l = 0$, with* $\lim_{r\to 0}(u(r) - ru'(r)) = 0$.

This is Teschl, (10.45)–(10.50) and Thm 10.8, in units with $\hbar^2/2m = 1$. For a function u continuously differentiable on $[0,\infty)$, the condition for $l = 0$ says $u(0) = 0$. It excludes $u = e^{-\eta r}$, which corresponds to the function $e^{-\eta|x|}/|x|$ in § 14.2. In the proof above, the bound $|u_p(r)| \leq \text{const} \cdot r$ did that work: it excluded the second solution, which near 0 is at least a multiple of r^{-l} in absolute value, a constant for $l = 0$.

Self-check 15.1

1. Use the recursion to find v for $n = 3$ and $l = 0$.
2. Theorem 15.1 gives only a lower bound on each eigenspace. Use the steps of the proof of Theorem 15.4 to show that $-\mathrm{Ry}/5$ is not an eigenvalue of H, and that the level $n = 3$ has no tenth independent eigenvector.

15.2 The degeneracy n^2

The level $-\mathrm{Ry}/n^2$ holds the states of angular momentum $l = 0, 1, \dots, n-1$, with $2l+1$ independent states for each l and $\sum_{l=0}^{n-1}(2l+1) = n^2$ in all. Spectroscopy names $l = 0, 1, 2, 3$ by the letters s, p, d, f, so the level $n = 2$ holds one $2s$ state and three $2p$ states, and the level $n = 3$ holds $3s$, $3p$ and $3d$, with $1 + 3 + 5 = 9$ states.

Rotations explain part of this. Each $U(R)$ commutes with multiplication by V, which depends only on $|x|$. By the self-check of § 13.1 on $\mathcal{S}(\mathbb{R}^3)$ and the first and third claims of Lemma 14.1, each $U(R)$ maps $D(H_0)$ into itself and commutes with H_0. So the eigenspaces of H are kept by the rotations. Each is a sum of whole multiplets (Hall, p. 397), as *Quantum Mechanics I*, Cor. 7.5, leads one to expect of a Hamiltonian that commutes with rotations. That forces the $2l+1$ states of one l, and nothing more. That the angular momenta $l = 0, 1, \dots, n-1$ share one level is not forced by rotations. As in the oscillator of § 13.6, it comes from a larger symmetry. Optional: Hall, Thms 18.14 and 18.16, explains this symmetry using the Runge–Lenz vector. Nothing here depends on that construction.

Figure 15.3 sets the levels of hydrogen beside those of the oscillator of Example 13.13. The oscillator's level N holds one multiplet of each $l = N, N-2, \dots \geq 0$, and hydrogen's level n one of each $l = 0, 1, \dots, n-1$. In both, rotations force the $2l+1$ states of each multiplet into one level, and do not force several multiplets to share it.

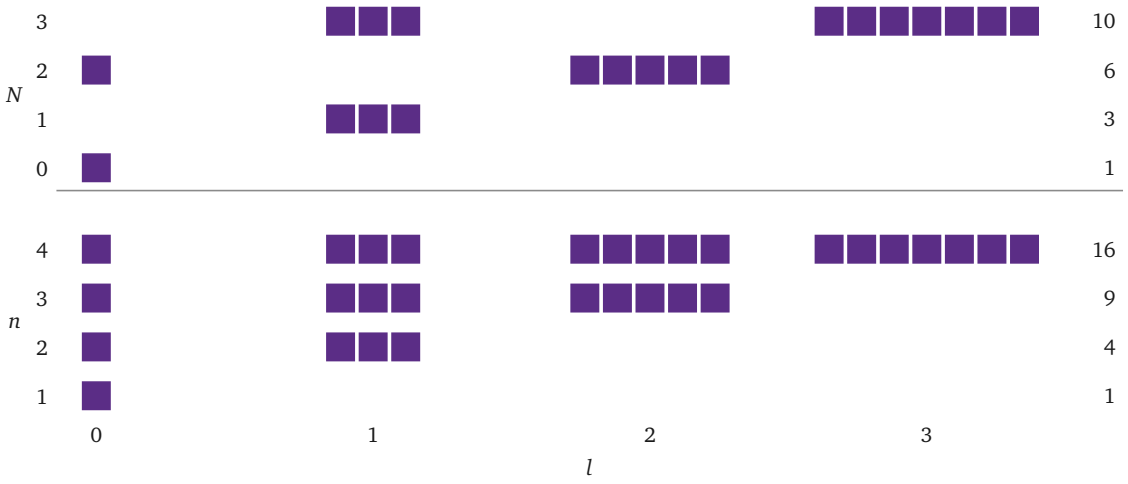


Figure 15.3: The states of the oscillator in space at its levels $N = 0, \dots, 3$ (above) and of hydrogen at its levels $n = 1, \dots, 4$ (below), one box for each independent state. A row is one level, and the rows are in order of energy, not to scale. A group of $2l+1$ boxes over the column l is one multiplet, and the number at the right of a row is the dimension of the level, $(N+1)(N+2)/2$ or n^2.

Self-check 15.2

1. How many independent states of H have an energy below $-\mathrm{Ry}/36$? Which angular momenta l occur at the level $n = 5$?

15.3 The Balmer lines

An atom in the level n can fall to a lower level n' and emit a photon, which carries the difference of the energies. This is a rule of the coupling to light, not derived here, as in the physics box of Lecture 13. A photon of energy E has the wavelength $\lambda = 2\pi\hbar c/E$, with the speed of light c.

Example 15.6 (The Balmer lines)**.**

1. *The energies.* A fall from $n \geq 3$ to $n' = 2$ emits $\mathrm{Ry}(1/4 - 1/n^2)$. For $n = 3$ this is $5\mathrm{Ry}/36 \approx 1.89\,\mathrm{eV}$.
2. *The wavelengths.* With the *Rydberg constant* $R_\infty = \mathrm{Ry}/2\pi\hbar c \approx 1.0974 \times 10^7\,\mathrm{m}^{-1}$,

$$\frac{1}{\lambda} = R_\infty\Big(\frac{1}{4} - \frac{1}{n^2}\Big).$$

   For $n = 3, 4, 5, 6$ this gives 656.1, 486.0, 433.9 and 410.1 nm, four lines in the visible: red, blue-green and two violet.
3. *The measurement.* The lines of hydrogen lie at 656.3, 486.1, 434.0 and 410.2 nm (NIST, in air). Each prediction is 0.03% short. The proton moves too. With the reduced mass $mm_p/(m + m_p)$ in place of m, the wavelengths grow by 0.05%. The air shortens them by 0.03%.

Figure 15.4 shows the levels, their degeneracy and these four lines together. The lines crowd because the levels crowd towards 0. Above 0 lies the continuous spectrum.

Self-check 15.3

1. A fall from $n = 2$ to $n' = 1$ emits the Lyman-α line. Find its wavelength.

The physics

Balmer found the rule $1/\lambda \propto \frac{1}{4} - 1/n^2$ in 1885 from four lines measured by Ångström. Here it follows from the operator H with its domain, and R_∞ from $\hbar$, m, e, ε_0 and c. The intrinsic spins of the electron and the proton, left out here, split each line further. The electron's spin couples to its orbital angular momentum, as in the doublet of sodium (*Quantum Mechanics I*, §13.6). The proton's spin splits the ground state into the two levels of the 21 cm line of *Quantum Mechanics I*, Lecture 10. That splitting is proportional to the density of the electron at the proton, $|\psi_1(0)|^2 / \|\psi_1\|^2 = 1/\pi a_0^3$.

15.4 What a continuous quantum system is

This course asked the four questions of *Quantum Mechanics I* for a particle, and the continuum changed each answer. In this course a continuous quantum system is, above all, a particle whose position ranges over a continuum. Its unbounded observables need domains. Its readings are given by spectral measures, and the distribution of a reading may have no point weights.

States. A state is the ray of a unit vector in a space of infinite dimension, L^2 of an interval, of the line or of space. Its vectors are classes of functions, and a position has no eigenvector in it. A scattering solution, which is $e^{ikx} + re^{-ikx}$ on the left of a potential, is not square-integrable, so it is not a state. For the potentials of Lecture 11, an incoming free packet φ gives the actual state $\Omega_{\mathrm{in}}\varphi$, which at early times moves like φ under the free evolution

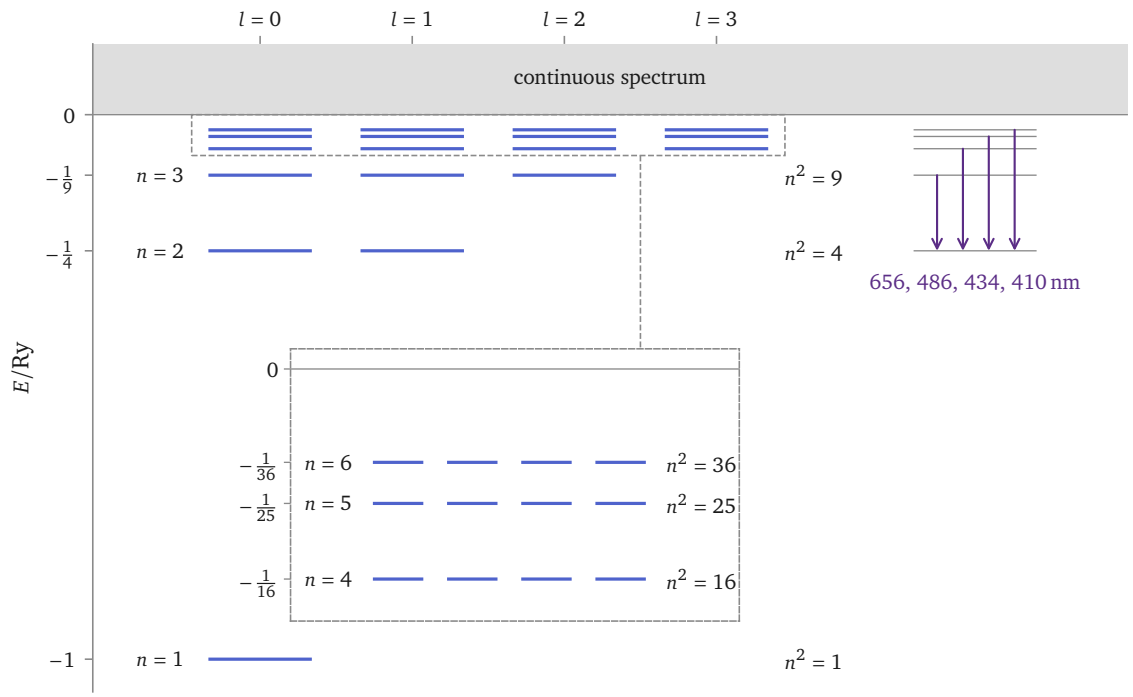


Figure 15.4: The levels $-\mathrm{Ry}/n^2$ of hydrogen for $n \leq 6$. A level appears in the columns $l \leq n-1$, and the columns $l > 3$ are not drawn. Beside a level is its degeneracy. The inset magnifies the dashed box below 0, which holds the levels $n = 4, 5, 6$, on its own energy scale. The arrows, in a strip of their own, are the four Balmer lines of Example 15.6, from $n = 3, 4, 5, 6$ to $n' = 2$. They show only the energies and say nothing about l. The grey band $E \geq 0$ is the continuous spectrum.

(Theorem 11.7). On the line the space is not a choice. Every strongly continuous irreducible Weyl pair, at a fixed $\hbar$, is the line's, up to a unitary map (Theorem 9.5).

Observables. An observable is a self-adjoint operator, with its domain. A symmetric operator is not enough. The momentum P_D in a box is symmetric and not self-adjoint, and its self-adjoint versions P_θ give different readings (Lecture 3). Nor is a formula enough. Hydrogen's rule $-(\hbar^2/2m)\Delta - C/|x|$ is self-adjoint on $D(H_0)$ with the least energy $-\mathrm{Ry}$, and its other self-adjoint extensions from the smooth functions that vanish near the origin each have a level below $-\mathrm{Ry}$ (§ 14.2). The spectrum of a self-adjoint operator is the set of numbers with approximate eigenvectors (Theorem 6.3). It is discrete in the box and the oscillator, continuous for the free particle, and has both parts in the finite well and in hydrogen. Figure 15.5 collects these spectra.

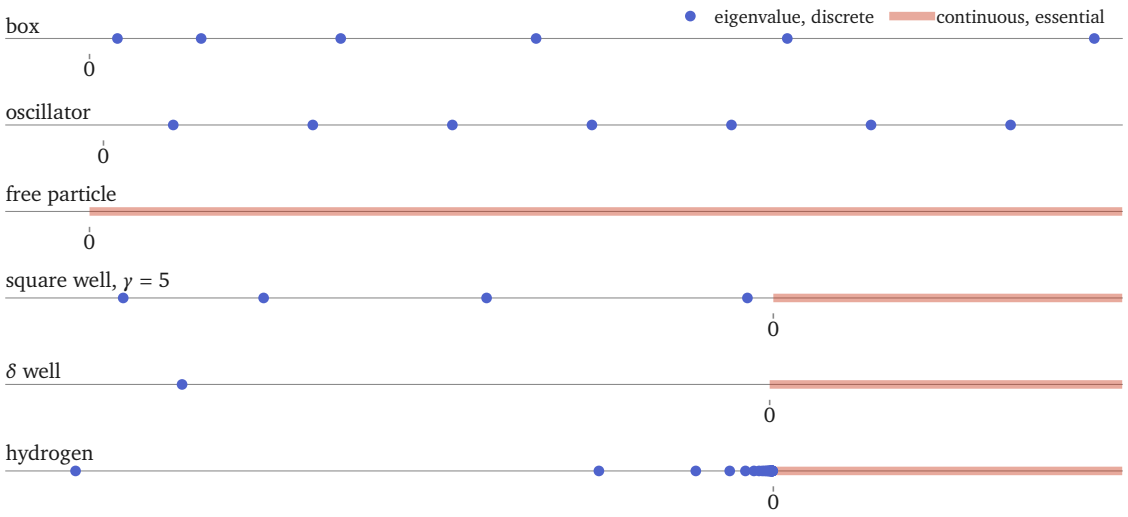


Figure 15.5: The energy spectra of the course, one per row, with the marks of Figure 6.2. The unit of energy differs from row to row, and only the mark 0 is common: E_1 for the box, $\hbar\nu$ for the oscillator, V_0 for the well, $m\alpha^2/2\hbar^2$ for the δ well and Ry for hydrogen.

Measurement. A reading asks whether an observable A lies in an interval I, and the spectral measure gives its probability (Postulate 5.6). If a yes has positive probability on the state of a unit vector ψ, then after a yes the state is the ray of $E_A(I)\psi$. A sharp reading of position on an interval can leave a state whose energy or momentum has an infinite mean square, as in the box (§ 2.6) and on the line (Example 5.7). For a packet coming in from the left, whose free momenta lie so near $\hbar k$ that $|t(p/\hbar)|^2$ hardly varies on them, $|t(k)|^2$ is approximately the late-time probability of being read on the right (Theorem 11.8).

Evolution. Evolution is a strongly continuous unitary group, and its generator is self-adjoint (Stone, Theorem 7.8). So self-adjointness is needed twice. Only a self-adjoint operator is the integral of a spectral measure, which gives its readings (Theorem 5.4). Only a self-adjoint operator, divided by $\hbar$, generates a strongly continuous unitary group. Every state evolves, even one outside $D(H)$. The map $t \mapsto U_t\psi$ satisfies the Schrödinger equation, in norm, exactly when $\psi \in D(H)$ (Theorem 2.2 in the box, Proposition 7.3 in general). A unit vector lies in $D(H)$ exactly when its energy readings have a finite mean square (§ 1.7 and Theorem 5.4). For a potential on the line, the states of $\mathcal{H}_{\text{pp}}$ stay near the well, and every state $\Omega_{\text{out}}\chi$ leaves every bounded interval (Theorem 12.6).

Quantum Mechanics III: Foundations takes up what this course left out: states that are density operators on an infinite-dimensional space, readings more general than spectral measures, and the question of what an observation is.

Hydrogen

States: the levels $-\text{Ry}/n^2$ have the eigenvectors $p(x)v(|x|)e^{-|x|/na_0}$, $p \in \mathcal{V}_l$, $l \le n-1$ (Theorem 15.1), and these span each eigenspace (Theorem 15.4). *Observables:* the energy $H = H_0 - C/|x|$, self-adjoint on $D(H_0)$ (Theorem 14.2), has the spectrum $\{-\text{Ry}/n^2\} \cup [0,\infty)$, and each level holds exactly n^2 independent states (Theorem 15.4). *Measurement:* a reading of the energy in an interval that misses $\{-\text{Ry}/n^2\} \cup [0,\infty)$

answers yes with probability 0 (Theorem 15.4, and Theorem 6.4, cited).

Summary.

- *The levels of hydrogen*: $-\mathrm{Ry}/n^2$, each with exactly n^2 independent eigenvectors, and no other eigenvalue (Theorems 15.1 and 15.4).
- *The degeneracy*: the level $-\mathrm{Ry}/n^2$ holds the angular momenta $l = 0, 1, \ldots, n-1$. Rotations force the $2l+1$ states of each l, and not that the l share one level (§ 15.2).
- *The Balmer lines*: $1/\lambda = R_\infty(1/4 - 1/n^2)$, within 0.03% of the measured lines (Example 15.6).

The end of the course. Quantum Mechanics III: Foundations recalls these answers and asks what they leave open.

Exercises. Exercise Sheet 15.

Self-check answers.

15.1 1. Here $c_1 = -2/3a_0$ and $c_2 = c_1(-2)/(2\cdot 3\cdot 3a_0) = 2/27a_0^2$, so $v = 1 - 2r/3a_0 + 2r^2/27a_0^2$. 2. For $E = -\mathrm{Ry}/5$, $\eta = 1/\sqrt{5}\,a_0$. Then $\eta(k+l+1) - 1/a_0 \neq 0$ for all integers $k, l \geq 0$, since $\sqrt{5}$ is not an integer. So no series stops, the fourth step gives $u_p = 0$ for every p, and the fifth gives $\psi = 0$. At $n = 3$, the fourth step gives $u_p = 0$ for $l \geq 3$, and the third gives $u_p = \beta(p)u_{3l}$ for $l \leq 2$. For bases of $\mathcal{V}_0$, $\mathcal{V}_1$ and $\mathcal{V}_2$ there are $1+3+5 = 9$ numbers $\beta(p_{l,i})$. By the fifth step, an eigenvector with all these numbers zero is zero. Ten independent eigenvectors would have a nonzero combination with all nine numbers zero, so there is no tenth.

15.2 1. Since $-\mathrm{Ry}/n^2 < -\mathrm{Ry}/36$ exactly when $n < 6$, these are the levels $n = 1, \ldots, 5$, with $1+4+9+16+25 = 55$ states. At $n = 5$, $l = 0$, 1, 2, 3 and 4.

15.3 1. $1/\lambda = R_\infty(1 - \frac{1}{4})$, so $\lambda \approx 121.5\,\mathrm{nm}$, in the ultraviolet.

References

Teschl, *Mathematical Methods in Quantum Mechanics*, Thms 10.2 and 10.3 (pp. 222–223), (10.45)–(10.50) and Thm 10.8 (pp. 229–230), Thm 10.9 (pp. 231–233). Hall, *Quantum Theory for Mathematicians*, Thms 18.3 and 18.4 (pp. 397–400), Cor. 18.5 (p. 401), Thms 18.14 and 18.16 (pp. 406–407). NIST Atomic Spectra Database, lines of H I. Teschl, Thm 10.9 (with his Thms 10.2, 10.3 and 10.8), obtains the negative eigenvalues of Theorem 15.4 and their multiplicities by separating the angular variables first. Hall, Thm 18.4 and Cor. 18.5, sketches the same route and cites the discreteness of the negative spectrum.